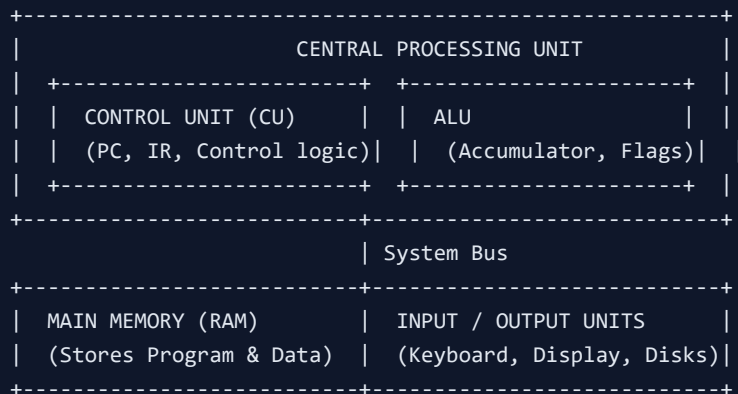


3-Mark Questions & Answers — MPCA Unit 1: Overview of Computer Architecture & Organization

Q1. Explain the Von Neumann model with a neat diagram.

The **Von Neumann Model** describes a computer system based on stored-program concepts, where instructions and data reside in a single shared memory.



- **Control Unit (CU):** Fetches instructions and directs operations.
- **ALU:** Performs arithmetic and logic computations.
- **Memory Unit:** Holds both program instructions and working data sequentially.

Q2. Explain any three addressing modes with examples.

1. **Immediate Addressing Mode:** The operand is specified directly within the instruction itself.
 - Example: `MOV AX, 0005H` (Loads value `0005H` directly into register `AX`).
2. **Register Addressing Mode:** The operand is stored inside a CPU register.
 - Example: `MOV AX, BX` (Copies 16-bit contents of `BX` into `AX`).
3. **Direct Addressing Mode:** The 16-bit offset address of the operand in memory is specified directly.
 - Example: `MOV AX, [2000H]` (Loads contents of memory location `DS:2000H` into `AX`).

Q3. Differentiate between CISC and RISC architecture.

Feature	CISC (Complex Instruction Set)	RISC (Reduced Instruction Set)
Instruction Count	Large set (300+ complex instructions).	Small set of simple, uniform instructions.
Execution Time	Variable (multiple clock cycles per instruction).	Fixed (single clock cycle per instruction).
Memory Access	Operations allowed directly on memory operands.	Load/Store architecture only (operands must be in registers).
Examples	Intel x86 (8086, Core i9), AMD Ryzen.	ARM, MIPS, RISC-V.

Q4. Explain the evolution of computers (generations) briefly.

1. **1st Gen (1940-1956):** Vacuum tubes; massive physical size; machine language; magnetic drums.
2. **2nd Gen (1956-1963):** Transistors; assembly & high-level languages (FORTRAN); magnetic core memory.
3. **3rd Gen (1964-1971):** Integrated Circuits (ICs); keyboards & monitors; operating systems introduced.
4. **4th Gen (1971-Present):** Microprocessors (VLSI/ULSI); personal computers (PCs); high-speed memory.
5. **5th Gen (Present & Future):** Artificial Intelligence, parallel processing, quantum computing, neural hardware.

Q5. What are the different types of instruction formats based on operand count?

1. **Three-Address Instructions:** Specify 2 source operands and 1 destination operand.
- Syntax: `ADD R1, A, B` (Meaning: `R1 = A + B`)
2. **Two-Address Instructions:** Specify 1 destination/source operand and 1 source operand.
- Syntax: `ADD A, B` (Meaning: `A = A + B`)
3. **One-Address Instructions:** Use an implicit **Accumulator (AC)** register.
- Syntax: `ADD B` (Meaning: `AC = AC + B`)
4. **Zero-Address Instructions:** Operate implicitly on a **Stack** data structure.
- Syntax: `ADD` (Pops top two stack items, adds them, and pushes result back onto stack).